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Pollination and Fertilization.

Pollination is carried out by wind and occurs at 3 celled stage(a prothallial cell, a large tube cell and a small generative cell). Pollen grains gets lodged in the pollen chamber after pollination. The generative cell divides into a stalk and a body cell. The body cell divides to produce two large multiciliated antherozoids or sperms. During fertilization, one of the male gamete or multiciliated antherozoid fuses with the egg of the archegonium to form a diploid zygote (2n). The endosperm is haploid. The interval between pollination and fertilization is 4- 6 months. The zygote undergoes mitotic division and develops into embryo. The ovule is transformed into seed. The seed has two unequal cotyledons. Germination is hypogeal. The life cycle shows alternation of generations (Figure 2.47).

2.6.5 Pinus

Class – Coniferopsida

Order – Coniferales

Family –Pinaceae

Genus - *Pinus*

Pinus is a tall tree, looks conical in appearance and forms dense evergreen forest in the North temperate and sub-alpine regions of the world. They mostly grow in high altitudes (ranging from 1,200 to 3,000 metres). Some species of this genus include, *Pinus roxburghii, P. wallichiana, P. gerardiana* and *P. insularis.*

External features

The plant body is sporophyte and is differentiated into root, stem and leaves. The main stem is branched. The branches are dimorphic with long and short branches (Figure 2.48).



Figure 2.48: *Pinus* Habit

Root

Tap root system is found in *Pinus*. The root hairs are not well developed and the roots are covered with fungal hyphae called mycorrhizae.

Stem

The stem is cylindrical, erect, woody and branched. The branches are monopodial. The branches are of two types.

(i) Long shoots or branches of unlimited growth, (ii) Dwarf shoot or branches of limited growth

(i) Long shoots or branches of unlimited growth

The long shoot is present on the main trunk the apical buds grow indefinitely, They shorten gradually towards the tip, thus providing a pyramidal appearance to the tree. These branches bear scale leaves only.